

baseline *real-time* *target machine*

Size S

User Manual



v2.2

speedgoat
real-time simulation and testing

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1 Introduction

The Baseline real-time target machine with an Intel® 2.0 GHz quad core CPU and optional FPGAs is the MathWorks® and Speedgoat entry-level solution to real-time simulation and testing using MATLAB® & Simulink®. This light, rugged and high-performance target machine is the perfect solution for running complex models requiring just a few hundred I/O channels.

It can be used as a real-time development platform in a broad range of application areas and industries such as the aerospace, automotive, machinery, electricity, consumer electronics', medical devices' and robotics' industries.

With a rugged enclosure and M12 connectors, this target machine can readily be moved around: it can be used for office, in-vehicle and even offshore operations. The enclosure is designed to withstand high levels of shock and vibration, and has a passive heatsink for silent fanless operation. The Baseline has capacity for up to six I/O modules covering the latest FPGAs (Xilinx® Artix®-7 and Kintex®-7) and a wide range of I/O connectivity.

1.1 Built to Your needs, Made for Simulink

MathWorks software products are optimized to guarantee the best real-time performance, workflow integration and usability with Simulink Real-Time™ and Speedgoat target machines. The latest Simulink and Speedgoat technologies are supported with this platform.

1.2 Concurrent Real-Time Application Execution

Leverage the full potential of latest generation Intel CPU and Xilinx FPGAs. MathWorks software, including Simulink Real-Time, MATLAB Coder™, Simulink Coder™, and HDL Coder™, and Speedgoat real-time target machines and FPGA-based I/O modules allow concurrent execution of your real-time application created from Simulink models on multiple cores, target machines, and FPGAs.

1.3 FPGA Co-Execution

Run parts of your Simulink models on Simulink-Programmable FPGAs at closed-loop sample rates of up to several MHz. Cost-effective and powerful Xilinx Kintex 7 FPGA-based I/O modules from Speedgoat, combined with HDL Coder from MathWorks, and x86/FPGA co-execution functionality provided by Simulink Real-Time, offer you a seamless workflow to automatically generate, run, verify, and validate parts of your real-time application on one or several FPGAs.

1.4 High-Grade Chassis

The enclosure of the Baseline real-time target machine is manufactured from high-grade aluminum which is ideal as it is strong, very light, corrosion resistant, conducts heat and is non-magnetic. High-grade aluminium also makes the enclosure attractive.

2 Safety Guidelines

2.1 Intended Use

The Baseline real-time target machine has been designed, developed, and manufactured for use in industrial environments. When used in an environment where there are serious risks or hazards that could lead to death, injury, serious physical damage or loss of any kind, users must observe exceptionally stringent safety precautions. In particular, such risks and hazards include the use of these devices to monitor nuclear reactions in nuclear power plants, their use in flight control or flight safety systems as well as in the control of mass transportation systems, medical life support systems or weapons' systems.

2.2 Protection Against Electrostatic Discharge

Electrical components that can be damaged by electrostatic discharge (ESD) must be handled accordingly.

2.3 Packaging

2.3.1 Electrical Components with Housing

These do not require special ESD packaging but must be handled appropriately.

2.3.2 Electrical Components Without Housing

These must be protected by ESD-suitable packaging.

2.4 Guidelines for Appropriate ESD Handling

2.4.1 Electrical Components with Housing

- Do not touch the connector contacts on the connected cables.
- Do not touch the contact tips on the circuit boards.

2.4.2 Electrical Components Without Housing

The following applies in addition to the points listed under the "Electrical Components with Housing" section above:

- Any persons handling electrical components or devices featuring installed electrical components must be grounded
- Components should only be handled on the narrow side or front plate
- Components should be stored in a suitable medium (ESD packaging, conductive foam, etc.)
- Metallic surfaces are not suitable storage surfaces
- Components should not be subjected to electrostatic discharge (for example, through the use of charged plastics)
- Ensure a minimum distance of 10 cm from monitors and TV sets
- Measurement devices and equipment must be grounded
- Measurement probes on potential-free measurement devices must be discharged on sufficiently grounded surfaces before taking measurements

2.4.3 Individual Components

- ESD-protective measures for handling individual components are thoroughly integrated at Speedgoat (conductive floors, footwear, wrist straps and so on).
- These increased ESD-protective measures for individual components are not necessary for customers handling Speedgoat products.

2.5 Policies and Procedures

Electronic devices are never completely failsafe. If the system fails, the user is responsible for ensuring that any other connected devices, for example, motors, are brought to a secure state.

Safety precautions relevant to industrial control systems (for example, the provision of safety devices such as emergency stop circuits, and so on) must be observed in accordance with applicable national and international regulations. The same applies for all other devices connected to the system, such as drives.

All tasks such as the installation, commissioning and servicing of devices are only permitted to be carried out by qualified personnel. Qualified personnel are those familiar with the transport, mounting, installation, commissioning and operation of devices who also have the appropriate qualifications (for example, IEC 60364). National accident prevention regulations must be observed.

Safety notices, connection descriptions (type plate and documentation) and limit values listed in the technical data must be read carefully before installation and commissioning and must be observed.

2.6 Transport and Storage

During transport and storage, devices must be protected against undue stress (mechanical loads, temperature, humidity, aggressive atmospheres, and so on).

2.7 Installation

- Installation must be performed according to the Speedgoat documentation for the real-time target machine.
- Devices must only be installed by qualified personnel without voltage applied. Before installation, voltage to the control cabinet must be switched off and prevented from being switched on again.
- General safety guidelines and national accident prevention regulations must be observed.
- Electrical installation must be carried out according to applicable guidelines (for example, line cross sections, fuses and protective ground connections).

2.8 Operation

2.8.1 Protection Against Contact with Electrical Parts

To operate, it may be necessary for certain parts to carry dangerous voltage levels over 42 VDC. Touching one of these parts can result in a life-threatening electric shock. This could lead to death, severe injury or damage to equipment.

Before turning on the Baseline real-time target machine, the housing must be properly grounded. Ground connections must be established even when testing/operating devices or the uninterruptible power supply for a short time!

Before turning on the device, all parts that carry voltage must be securely covered. During operation, all covers must remain closed.

2.8.2 Environmental Conditions – Dust, Humidity and Aggressive Gases

Use in very dusty environments should be avoided. Dust accumulation on the devices can affect functionality.

The presence of aggressive gases can also lead to malfunctions. When combined with high temperature and humidity, aggressive gases, for example, sulfur, nitrogen and chlorine components, can induce chemical reactions that can damage electronic components very quickly. Blackened cable ends on existing equipment indicate the presence of aggressive gases.

2.8.3 Viruses and Dangerous Programs

This system is subject to potential risk each time data is exchanged or software is installed from a data medium (for example, a USB memory drive), a network connection or the Internet. The user is responsible for assessing these dangers, implementing preventive measures such as virus protection and firewalls, etc., and making sure that software is only obtained from trusted sources.

3 Installation

Place your Baseline real-time target machine on a stable desk or install it on DIN rails using the DIN rail clips provided.



3.1 Monitor

You can connect one DisplayPort monitor to your target machine. The monitor is not mandatory, but it is recommended for kernel transfer and debugging operations.

For the first session with your new target machine, please connect a monitor. The target main screen provides helpful information:

- Simulink Real-Time version
- Number of CPU cores loaded
- Available memory
- TCP/IP configuration for communications between the development computer and the target machine

You can also use Target Scopes in the Simulink model to visualize signals and parameters at execution time. If you do not use a monitor during real-time operations, you could run the kernel in text mode, to improve performance.

If you do use a monitor, connect it before powering up the target machine. Make sure that the monitor is well connected and do not unplug it during model execution. This could have an impact on the application stability and in some particular cases, crash the system. If this problem occurs, you must restart the target machine.

3.2 Keyboard

You can use PS/2 and/or USB keyboards.

PS/2 keyboard: enter the BIOS, and use DOS commands or Simulink Real-Time commands

USB keyboard: enter the BIOS and use Simulink Real-Time commands

The keyboard is not mandatory for Simulink Real-Time.

To activate the command line interface once you have a target application running, type C on the attached keyboard. For a list of commands available in the Simulink Real-Time environment, please refer to the Simulink Real-Time documentation.

3.3 Power Supply

Connect the provided power supply to the power input at the rear of the target machine. For the technical specification of the power supply and voltage input range, please refer to chapter "5.1.3 Power Input". For more information about the power connector, refer to chapter "5.5 Power Connector".

3.4 Power On/Off

The system will power up as soon as the power supply is connected to the power input. When the power LED is blue, the system is correctly powered on.

The system usually takes a few seconds to load the Simulink Real-Time kernel stored in the target machine main drive. Once the kernel is loaded, kernel information will appear on the target screen.

To power off the system, unplug the power connector or press the green power button.

4 Software

4.1 Installation and Configuration

Refer to the [Software Configuration Guide](#) to complete the configuration of the real-time target machine. The Software Configuration Guide covers:

- MathWorks software and Speedgoat Library installation
- Ethernet communications' link
- Kernel configuration and transfer

4.2 Simulink Real-Time Environment

For help with the Simulink Real-Time environment, please refer to the MathWorks Documentation: www.mathworks.com/help/xpc

4.3 Driver Block Manuals

When you install the Speedgoat Library, the installer sets up help for the blocks in the MATLAB Help browser. To view the Speedgoat Help documentation, open the MATLAB Help browser and go to the home page. At the bottom-right of the home page, under Supplemental Software, click Simulink Real-Time - Speedgoat. The Speedgoat Help documentation opens in the current window.

PDF driver block manuals for all of your installed I/O modules are available for download in the Speedgoat Customer Portal on the Speedgoat website.

4.4 DOS Loader and Drivers

This Baseline real-time target machine is ready for use with the Simulink Real-Time kernel, which is loaded from a DOS installation. The DOS environment also allows USB kernel transfer so that the system can be updated with a new Simulink Real-Time kernel.

DOS version	freeDOS 1.0
USB driver	DOSUSB 2.0
USB kernel transfer tool	C:\AUTOEXEC.BAT prepared by Speedgoat

4.5 Simulink Real-Time Kernel

Speedgoat Real-Time target machines are optimized for use with Simulink Real-Time. By default, Speedgoat pre-installs the Simulink Real-Time kernel on your target machine. However, to work with Simulink Real-Time, a valid Simulink Real-Time license from MathWorks is required.

This real-time target machine is compatible with MATLAB release R2016a or later.

Kernel version	6.4 or newer
IP address	192.168.7.1
Port	22222
Subnet mask	255.255.255.0
Gateway	255.255.255.255
TCP/IP driver	INTEL_I210
USB Support	Disabled
Multicore CPU	Enabled
Graphics mode	Enabled

Further information on how to transfer the Simulink Real-Time kernel version for the respective MathWorks software release to your target machine and to change the IP settings of your target machine is available in the Software Configuration Guide provided with your target machine.

For more information about the Simulink-Real-Time software and drivers for the installed I/O modules, refer to the individual I/O module User Manuals and Simulink Driver Block Manuals. These are available at www.speedgoat.com/help.

4.5.1 Kernel Settings

Use the **Real-Time Kernel Configuration and Transfer** tool when a new kernel is required, for example, to change the target's IP address. More information about this tool is available at https://www.speedgoat.com/help/page/configuration/refentry_ref_target_setup

When the target driver and ethernet index are set manually, the following settings are required to define the top-left Ethernet port as the host-link.

Target driver	I210
Ethernet index	3

5 Technical Specifications

5.1 Baseline Real-Time Target Machine

The Baseline real-time target machine meets the specifications described in this manual and is designed, manufactured, assembled, and tested in Switzerland.

Product name	Baseline real-time target machine
Country of origin	Switzerland
Operating temperature	0 °C to +60 °C (32 °F to +140 °F) ExtTemp (itemID 107110) -40 °C to +85 °C (-40 °F to +185 °F)
Humidity	5 % to 90 %
Weight	2.6 kg

5.2 Enclosure

Dimensions	Width: 190 mm Depth: 190 mm Height: 80 mm
Fans	Fanless design. Passive cooling

5.3 External Power Supply

Manufacturer	Mean Well
Product	AC/DC switching adapter GST90A24
AC input	100-240 V AC, 1.3 A, 50-60 Hz
DC output	24V DC, 3.75 A max

Speedgoat will deliver the most suitable power supply for the given real-time target machine configuration.

Manufacturer	Mean Well
Product	AC/DC switching adapter GST220A24
AC input	100-240 V AC, 4.0 A, 50-60 Hz
DC output	24 V DC, 9.2 A max

5.4 Power Input

Input voltage range	8 to 36 VDC (12 V $\pm$ 33 % ... 30 V $\pm$ 20 %)
Power consumption (no modules installed)	15W

5.5 Mainboard

Revision type	REV 4
CPU	Intel Celeron® J1900, Quad Core, 2GHz, 2 MiB L2 Cache
CPU Option	Intel Atom™ E3845, Quad Core, 1.91 GHz, 2 MiB L2 Cache (ItemID 107110)
Chipset	Integrated
System memory	4 GiB DDR3 Non-ECC Bundle
Graphics (standard)	Intel HD Graphics, 688 – 854 MHz
Ethernet	4x Intel I210 PCI Express Gigabit Ethernet controllers
Serial ATA	1x mSATA 2.0
mPCIe slots	4x mPCIe slots
PCI bus clock	33/66 MHz
Accessible I/O ports	1x PS/2 keyboard 1x USB 3.0/2.0/1.1 2x USB 2.0/1.1 2x RS232 1x DisplayPort 4x Gigabit Ethernet ports (3x external RJ45, 1x internal FCI) 1x Ignition (with power and reset control)
BIOS	AMI BIOS
Dimensions	Double layer mini ITX Width: 170 mm Depth: 170 mm Height: 55 mm

5.6 SSD Storage

Manufacturer	Advantech
Product	SQF-SMSM4-32G-S9E
Size	32GB SSD
Options	128 GB SSD (ItemID 107251) 256 GB SSD (ItemID 107252)

5.7 Thermal Design

5.7.1 Passive Cooling

The Baseline real-time target machine is convection cooled by the heat sink surrounding the device. All the internal heat sources are thermally connected to the aluminum chassis.

There are no fans inside the machine. For heat dissipation, convection cooled systems require a constant airflow around the enclosure. Do not place this unit in a closed box with no air circulation.

When operating in very hot environmental conditions, the target machine can become very hot. However, the temperature may still be acceptable for the electronic components; for example, an automatic shutdown of the complete unit is only required if the CPU temperature exceeds 125 °C. For many other components, a shutdown is required if the temperature exceeds 85 °C.

In such extreme conditions, follow these instructions:

- Do not touch the chassis with bare skin
- Keep the machine away from heat-sensitive objects
- Install external ventilation to ensure airflow through the silver fins

For most applications, despite the machine getting warm, you will still be able to handle it without any risk of injury.

5.7.2 Over Temperature Shutdown Procedure

The internal temperature is measured in several locations. If an over temperature is detected, the supervision shuts down the system.

After a shutdown, please press the power button once. Solely cycling the power input will not restart the machine.

5.8 LED Status Indicators

5.8.1 Power

Power indicates the power state.

LED Behavior	Explanation
ON (Blue)	The machine is in ACPI power state S0 (Working) and the CPU temperature is below 100 °C

5.8.2 SSD

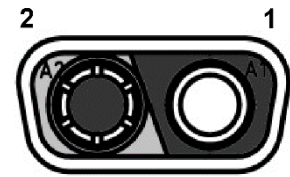
The SSD LED monitors the hard drive (SSD) activity.

LED Behavior	Explanation
Blinking (Blue)	The mSATA SSD is active.

5.9 Power Connectors

5.9.1 Power Input Connector

Pin	Description
1	Positive DC input voltage
2	Ground for DC input voltage



One of each of the components in the table below is required to assemble a complete mating connector:

Component	Item ID	Manufacturer/Supplier
Metal shell	970-009-040R011	NorComp/Mouser
Connector	302W2CPXX99A10X	CONEC/Mouser
High Power Contact M	131C11029X	CONEC/Mouser
High Power Contact F	132C11029X	CONEC/Mouser

5.9.2 5 V output pins

If the Baseline is delivered with an I/O module providing a 5 V output, the 5 V are provided by the mainboard through internal cabling. The power distribution switch on the mainboard has the following characteristics.

- Thermal and short-circuit protection
- 1 A continuous current
- 1.9 A current limit
- 0.6 ms typical rise time

All 5 V outputs are connected in parallel, meaning the current is split up between the outputs.

5.10 Standard I/O Connectors

The following I/O connectors can be used with Simulink Real-Time.

All the connectors are ESD protected.

5.10.1 Keyboard

Port	Supported Mode
Keyboard	Standard PS/2 keyboard input for DOS, BIOS and Simulink Real-Time

5.10.2 Video

Port	Signal Supported
DisplayPort	Supports resolutions up to 2560x1600 at 60 Hz Dual mode DisplayPort supports external passive DisplayPort to DVI, DisplayPort to VGA or DisplayPort to HDMI adapters

Only one monitor can be used at the same time. Simulink Real-Time requires a minimum 1280x1024x16-bit resolution standard VESA monitor.

5.10.3 Ethernet Controllers

Port	Bus Index ¹	Controller	Driver	Usage
FCI LAN	0	Intel I210	I210	industrial protocols
ETH 1	1	Intel I210	I210	industrial protocols
ETH 2	2	Intel I210	I210	industrial protocols
Host Link	3	Intel I210	I210	host link

5.10.4 Supported Functionality

Functionality	Host Link	ETH 1, ETH 2 & FCI-LAN
Host-target communication	x	
XCP Slave (for CANape and INCA)	x	
UDP (non real-time)	x	
Real-time UDP		x
EtherCAT Master		x
XCP Master over UDP		x

5.10.5 Serial I/O

The Baseline real-time target machine by default provides 2 x RS232 ports. Drivers for on-board serial communication are available in the `slrtlib` library.

Port	Address	IRQ line	Standards supported
COM 1	0x3F8	4	RS232
COM 2	0x2F8	3	RS232

Supports transfer rates for on-board serial ports up to 115.2 kBaud. For faster baud rates with RS232 and RS422/RS485 support, dedicated I/O modules are available from Speedgoat.

5.10.6 USB Support

USB ports are available for the following use cases:

- USB keyboard
- External USB storage device
- USB UVC webcams. USB 3.0 UVC webcams are supported starting from Matlab R2017a.

Note that for these features to work, USB support must be enabled in the kernel options of the **Real-Time Kernel Configuration and Transfer** tool. Note also that leveraging USB support may introduce additional latencies.

Port	Supported Mode
USB 2.0	2 ports for Simulink Real-Time supporting USB 1.1 and USB 2.0
USB 3.0	1 port for Simulink Real-Time supporting USB 1.1, USB 2.0 and USB 3.0

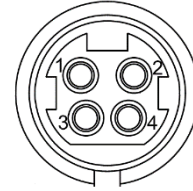
¹ The Bus index indicates the order in which the ethernet controllers will be enumerated on the PCI Bus. For the Bus/Slot configuration of your real-time target machine, refer to the technical reference manual.

5.10.7 Ignition: Remote Control

The ignition port enables connection to an external ignition key switch to start and stop the machine. It also allows external power- and reset-buttons to be connected.

To use this function, switch it on in the BIOS settings. The default configuration is off.

Pin	Signal
1	Ignition Input
2	Power Button Input
3	Reset Button Input
4	GND (Power Connector 0V)



The Mini-DIN connector counterpart with interlock is available from Speedgoat or directly from Mouser (806-KPPX-4P).

5.10.8 Using the Ignition Input

To activate the target machine, connect the ignition input pin 1 to GND pin 4 using a key switch or toggle switch. To deactivate the machine, leave pin 1 unconnected.

Optionally, the active level for the ignition signal can be inverted by changing a jumper on the mainboard. Please ask Speedgoat for more details.

5.10.9 Using the Power Button Input

To control the machine with a remote button, simply connect pin 2 and pin 4 to an open push button.

5.10.10 Using the Reset Button Input

To reset the machine with a remote button, simply connect pin 3 and pin 4 to an open push button. A reset button is generally not required, as pressing the power button twice has the same effect.

6 BIOS configuration

BIOS Vendor	American Megatrends
Version	BT9A3B76.21A
Build date	07/18/2017

6.1 Recommended Speedgoat BIOS settings

Each Baseline real-time target machine is optimized to provide best real-time performance with Simulink Real-Time. We therefore strongly recommend that the BIOS settings below are not changed. Doing so may reduce real-time performance and change the IRQ assignments.

6.2 Advanced Tab

BIOS Setting	Default Value
PC Health Status	
CPU SmartFan	Disabled
WatchDog Configuration	
WatchDog1 function	Disabled
Super IO Configuration	
Serial Port 1 Config.	
Serial Port	Enabled
Change Settings	Auto
Serial Port 2 Config.	
Serial Port	Enabled
Change Settings	Auto
CPU Configuration	
Intel Virtualization Technology	Disabled
EIST	Disabled
SATA Configuration	
Serial-ATA (SATA)	Enabled
Serial-ATA Port 0	Enabled
Serial-ATA Port 1	Enabled
Network Stack Configuration	
Network Stack	Disabled
CSM Configuration	
CSM Support	Enabled
Boot option filter	UEFI and Legacy
Network	Do not launch
Storage	Legacy only
Video	Legacy only
Other PCI devices	UEFI only
USB Configuration	
Legacy USB Support	Disabled
USB Mass Storage Driver Support	Disabled
Security Configuration	
TXE HMRFP0	Disabled

6.3 Chipset Tab

BIOS Setting	Default Value
North Bridge	
Intel IDG Configuration	
Integrated Graphics	Enabled
LCD Control	
Primary IGFX Boot Display	VBIOS Default
LCD Panel Type	Type 3
LVDS Support	Disabled
Memory Configurations	
Memory ReMap	Disabled
South Bridge	
Azalia HD Audio	
Audio controller	Disabled
PCI Express Configuration	
PCI Express Port 0-3	Enabled
Speed	Auto
Restore AC Power Loss	Last State
Intel® I210 controller	Enabled
Global SMI	Enabled
Legacy USB2 SMI	Disabled
SW SMI	Disabled
Sleep SMI	Disabled
Periodic SMI	Disabled
XHCI SMI	Disabled
SW SMI Timer	Disabled
Ignition	Off

6.4 Boot Tab

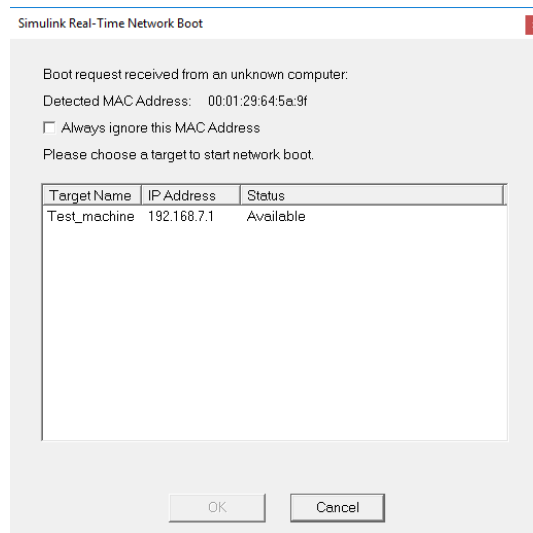
BIOS Setting	Default Value
Setup Prompt Timeout	1
Bootup NumLock State	ON
Boot Option #1	SQF-SMSM4-32G-S9E
Hard Drive BBS Priorities	
Boot Option #1	SQF-SMSM4-32G-S9E

7 Kernel Recovery

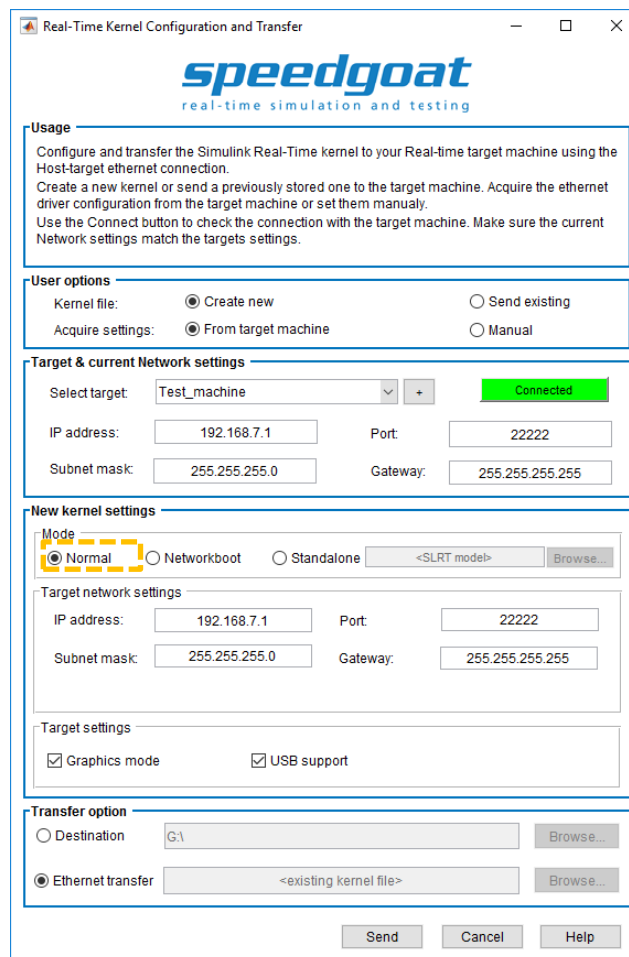
The real-time target machine's kernel must be recovered when the host cannot establish a connection with the target. Incorrect ethernet settings defined when the new kernel is loaded to the target machine are the most frequent cause of this issue.

1. Open the **Real-Time Kernel Configuration and Transfer** tool and select the **Networkboot** mode
 - a. **User options:** select **Manual**
 - b. **New kernel settings – Mode:** select the **Networkboot** mode.
 - c. **New kernel settings – Target network settings:** Target driver: **I210**, Ethernet Index: **3**

2. Click the **Create** button. This will create a network boot server
3. Boot the Baseline and press **delete** on the keyboard connected to the Baseline until the **BIOS** window appears. In the **BIOS** menu, select the following:
 - a. **Advanced>CSM Configuration>Network = Legacy only**
 - b. Save the changes and exit
4. While rebooting, press **delete** on the keyboard connected to the Baseline until the BIOS window appears. In the BIOS menu, select the following:
 - a. **Boot>Boot Option #1 = IBA GE Slot 1000 v1410**
 - b. Save changes and exit
5. The Baseline will now try to boot over the network. During this process, the network boot server on your host will open a window where you must select the target.



6. Once booted, open the **Real-Time Kernel Configuration and Transfer** tool and create a kernel in normal mode and then transfer it over the ethernet connection.



7. The Baseline will reboot again. Press **delete** on the keyboard connected to the Baseline until the BIOS window appears. In the BIOS menu, select the following:
- Advanced>CSM Configuration>Network** = Do not launch
 - Save changes and exit.
8. The Baseline will now boot from the SSD with the new kernel.

8 Standards and Certifications

This target machine meets or exceeds the following standards:

CE mark	The CE mark certifies that all harmonized EN standards for the applicable directives have been met for this Speedgoat product.
EMC directive	These devices meet the requirements of EC directive EN 61000-6-4 and EN 61000-6-2.
Low voltage directive	These devices meet the requirements of EC directive EN60950-1.

9. Legal

As used herein, the term "Seller" shall mean Speedgoat GmbH, and term "Buyer" shall mean the person, firm or corporation executing a purchase order for "goods", sold by Seller (hereinafter "Products").

9.1 Limited Warranty

Seller warrants that the Products delivered hereunder shall be free from defects in workmanship and material under normal use and wear in accordance with Seller's instructions and specifications for a period of twenty-four (24) months from date of delivery to the Buyer, including component parts of Products sold as spare, replacement, maintenance or storage parts, which are also warranted for twenty-four (24) months from date of delivery, provided, however, in either case, that notice of any such defect is provided to Seller within thirty (30) days of its discovery by the Buyer. THE WARRANTY SET FORTH IN THIS SECTION SHALL BE IN LIEU OF ALL OTHER WARRANTIES, AND ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTY OF MERCHANTABILITY, FITNESS FOR PARTICULAR PURPOSE, AND FITNESS FOR ORDINARY PURPOSE USED OR PURPOSE INTENDED, ARE EXCLUDED. IN NO EVENT SHALL SELLER, ITS EMPLOYEES OR SUPPLIERS BE LIABLE, EITHER DIRECTLY OR BY WAY OF INDEMNIFICATION, TO BUYER OR ANY THIRD PARTY FOR (A) AN AMOUNT EXCEEDING THE PURCHASE PRICE OF THE PRODUCT IN QUESTION AND (B) ANY PUNITIVE, EXEMPLARY, SPECIAL, INDIRECT OR CONSEQUENTIAL LOSSES, DAMAGES OR INJURIES regardless of whether such claim is based upon delays in delivery or payment, breach of warranty, breach of contract, strict liability, negligence, or any theory now known or hereinafter adopted by legislation or by any court. Neither Seller nor its affiliates shall be liable for any damage or loss to exposure of Products and/or their packaging to the elements (including but not limited to rain, snow, sleet, sun, wind, floods, etc.); chemicals, corrosive solvents or soils; unauthorized or improper use, maintenance, storage or repair; due to any failure to follow Seller's manuals, warnings, notices or instructions; or due to any malfeasance, recklessness or negligence by Buyer, any employee or costumer of Buyer or any other third party.

EXCLUSIVE REMEDY: In any event, the Buyer's exclusive remedy hereunder is limited to the furnishing of replacement parts on an exchange basis, or, at the option of Seller, to the repair or replacement of defective Products or replacement parts at one of Seller's locations, but in either case only if the defective Product or part has been submitted to Seller during the period of warranty. The Buyer accepts and acknowledges that the foregoing allocation of risk is reflected in the purchase price.

The parties further agree that if any portion of the foregoing exclusion of damages is held to be voidable or void by reason of public policy or unenforceable for any other reason whatsoever, all remaining portions of the foregoing exclusion shall continue in effect.

THE WARRANTY SET FORTH ABOVE DOES NOT EXTEND TO: Any systems which have been damaged or rendered defective as a result of accident, misuse, or abuse; by the use of parts not manufactured, authorized or sold by Seller; by modification or as a result of service by anyone other than Seller; systems not containing original components or original replacement of components; damage during shipment, unless due to incorrect packaging by Seller; systems which fail or are damaged after delivery due to shipment, handling, storage, operation, use or maintenance in manner or environment not conforming to any published instructions or specifications issued by Seller.

In-warranty repaired or replacement parts or Products are covered by warranty only for the remaining unexpired portion of the original warranty period applicable to the repaired or replaced parts or Products. In other words, repair or replacement of Products or parts under warranty does not extend the original warranty period.

Products that are no longer part of the regular sales offering are considered EOL (end-of-life) and are repaired on a best-effort basis.

EXTENDED HARDWARE WARRANTY SERVICE

Extended Hardware Warranty Service is available as an option and must be purchased at the time the Products are purchased for which the warranty shall be extended.

The **Level One Hardware Warranty Service** extends the standard 24-month warranty period by 12 months resulting in a 36-month warranty period.

The **Level Two Hardware Warranty Service** extends the standard 24-month warranty period by 36 months resulting in a 60-month warranty period.

Hardware warranty terms exceeding the 60-month range are available on request.

9.2 Return

Buyer shall not return any Product without Seller's prior written consent. An R.M.A. (Return Material Authorization) number issued by Seller must accompany all returned material. An RMA number can be obtained by contacting the Seller's support department (support@speedgoat.com).

Within Warranty, Products returned and needing corrective repair are serviced at no-charge in accordance with the terms of Seller's Warranty policy.

Repairs on out of Warranty Products are performed at Buyer's expense.

Please pack the returned Products in their original shipping cartons, or in equivalent strong protective shipping cartons. Shipping costs from Buyer to Seller associated with warranty repairs or replacements shall be borne by the Buyer. Shipping costs for the return of repaired goods from Seller to Buyer shall be borne by Seller.

9.3 Systems Software Maintenance and support services

Delivery of Seller systems and hardware/software components by default include 12 months (1 year) of Systems Software Maintenance and Support Services.

Subscription to Systems Software Maintenance and Support Services includes access to Seller tools and driver software compatible with future releases of MathWorks software and professional technical support by phone and e-mail.

Subscription to Systems Software Maintenance and Support Services does not include free updates of existing custom implementations (FPGA bitstreams).

Software Maintenance and Support Renewal

For uninterrupted Systems Software Maintenance and Support Services in subsequent years Buyer may opt to renew its subscription annually to maintain its investment. Reinstatement if elapsed is possible on request, but incurs back maintenance charges of up to 6 months. Staying subscribed is the most cost-effective way to access latest advances and technical support.

9.4 Use of Speedgoat Software Including Tools and Drivers

LEGAL INFORMATION ABOUT THE USE OF SPEEDGOAT TOOLS AND DRIVERS: Speedgoat tools and drivers are optimized for hardware purchased from Seller and may be used only in conjunction with the hardware (serial no.) for which the tools and drivers were purchased for. Access to the Speedgoat tools and drivers is only available if the target machine component has active subscription to Systems Software Maintenance and Support Services.

Terms and Conditions for software components are defined in the Speedgoat End-User License Agreement (EULA).